University of Engineering & Technology

Lahore, Pakistan

Department of Computer Science

FYP Management System

A Desktop Application for Final Year Project Committee

Report

CS104 – Database Systems (Spring 2026)

Project – Mid Term Lab Examination

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Submitted To:	Sir Nazeef Ul Haq

Abstract

This report presents the FYP Management System — a desktop application built using C# Windows Forms and backed by a MySQL relational database. The primary purpose of this application is to help the Final Year Project (FYP) committee at the University of Engineering and Technology (UET) Lahore manage their administrative responsibilities in a more organised, reliable, and efficient manner.

Previously, the FYP committee relied entirely on Microsoft Excel spreadsheets to track students, projects, advisors, and evaluation marks. While spreadsheets are flexible tools for simple data entry, they are poorly suited to managing inter-related records across multiple categories simultaneously. Data redundancy, accidental deletions, formula errors, and difficulty in querying specific records were frequent problems. As the number of students and projects grew each year, these limitations became increasingly disruptive.

This system replaces those spreadsheets with a purpose-built software application that enforces data integrity through foreign key relationships, prevents duplicate entries, and allows the committee to retrieve structured information quickly. The application supports adding and managing students, advisors, and projects; forming student groups; assigning projects to groups; assigning advisors in defined roles; recording evaluation marks; and generating formatted PDF reports for official use.

The underlying database is `projectadb26`, provided by the course instructor. All database interactions are performed using hand-written SQL queries; the Entity Framework ORM is not used, as per the course requirements. The application strictly follows the principle of separating the user interface from database logic through a dedicated Data Access Layer.

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1. Introduction

1.1 Background

The Department of Computer Science at UET Lahore conducts Final Year Projects (FYP) every academic year. These projects are a core component of the undergraduate curriculum, requiring students to apply the knowledge and skills acquired throughout their degree to a substantial, original piece of work. The FYP committee is responsible for overseeing every stage of this process, from collecting project proposals from faculty members to evaluating final presentations.

Managing these responsibilities is a significant administrative challenge. The committee must maintain accurate records for hundreds of students, track which students belong to which group, monitor which projects have been assigned to which groups, ensure that each project has appropriate advisors, and record marks from multiple evaluation rounds. When these tasks were handled through Excel spreadsheets, there was no enforcement of rules such as a student belonging to only one group, or a project being assigned to only one group. Errors could go undetected for extended periods and correcting them was time-consuming.

1.2 What This Project Does

This project provides a dedicated software solution to address the shortcomings of the spreadsheet-based approach. A Windows Forms desktop application was developed in C# and connected to a MySQL relational database. By storing all data in a structured database, the system is able to enforce data integrity automatically through constraints, primary keys, and foreign keys. The application eliminates the risk of duplicate records and ensures that all relationships between entities — such as a student belonging to a group, or an advisor being linked to a project — are properly maintained.

The application allows a FYP committee member to perform the following tasks from a single unified interface:

- Add, view, edit, and delete student records.
- Add, view, edit, and delete advisor records.
- Add and manage project titles and descriptions.
- Form student groups and add students to them.
- Assign a project to a group.
- Assign advisors (main, co, and industry) to a project.
- Create evaluation types and record marks for each group.

- Generate PDF reports for the committee's official records.

Each of these operations is implemented with proper input validation and error handling, so the application communicates clearly with the user when something goes wrong rather than crashing or silently storing incorrect data.

1.3 Who Uses This System

This system is intended exclusively for members of the FYP committee at UET Lahore. It is a desktop application that runs on the Windows operating system and is designed for use on computers maintained by the department. Students do not interact with this application directly; they are managed as records within the system. The application assumes that the person using it is authorised and familiar with the committee's administrative workflow, so no user login or role-based access control has been implemented in this version.

2. Requirements

2.1 What the User Needs

The FYP committee member is the primary user of the system. Based on an analysis of the committee's existing Excel-based workflow and the course project specification, the following user needs were identified:

- Maintaining a complete and accurate list of all students with their personal information.
- Maintaining a list of all advisors along with their academic designations and contact details.
- Managing a catalogue of FYP project titles and descriptions.
- Creating groups of students and tracking group membership.
- Assigning exactly one project to each student group.
- Assigning up to three advisors to each project, each in a distinct role (Main Advisor, Co-Advisor, Industry Advisor).
- Recording evaluation marks for each group across multiple evaluation events.
- Generating printed PDF reports that can be shared or archived officially.

3. Database Design

3.1 Tables in the Database

3.1.1 Lookup Table

The `lookup` table is one of the most important tables in the schema because it serves as the single source of truth for all fixed categorical values used throughout the system. Instead of storing string values such as “Male”, “Female”, “Professor”, or “Main Advisor” directly in the rows of other tables — which would make queries inconsistent if spellings ever varied — the system stores only the integer ID of each value and joins back to the `lookup` table when the human-readable text is needed for display. This design makes the database more robust and consistent.

The categories stored in this table include `GENDER` (for classifying persons), `STATUS` (for group membership status), `DESIGNATION` (for advisor academic ranks), and `ADVISOR_ROLE` (for the three advisor roles assigned to projects).

Table 3.1: Lookup Table Data

Id	Value	Category
1	Male	GENDER
2	Female	GENDER
3	Active	STATUS
4	InActive	STATUS
6	Professor	DESIGNATION
7	Associate Professor	DESIGNATION
8	Assistant Professor	DESIGNATION
9	Lecturer	DESIGNATION
10	Industry Professional	DESIGNATION
11	Main Advisor	ADVISOR_ROLE
12	Co-Advisor	ADVISOR_ROLE
14	Industry Advisor	ADVISOR_ROLE

3.1.2 Person Table

The `person` table is the central table for storing personal information shared by both students and advisors. Rather than creating two separate tables that each duplicate fields like first name, last name, email, and contact number, the schema uses a single `person` table as a parent from which both the `student` and `advisor` tables inherit. This is a well-established pattern in relational database design sometimes called table inheritance or shared primary key inheritance. The `Gender` column references the `lookup` table, ensuring that only valid gender values can be stored.

3.1.3 Student Table

The `student` table stores the registration number of each student. It shares its primary key with the `person` table through a foreign key relationship, meaning that a row in the `student` table can only exist if a corresponding row with the same `Id` already exists in the `person` table. This guarantees that every student also has a complete personal record, and it avoids duplicating the common personal fields in both tables. When a student is added, the application first inserts a row into `person`, retrieves the auto-generated `Id` using `LAST_INSERT_ID()`, and then inserts into `student` using that same `Id`.

3.1.4 Advisor Table

The `advisor` table follows the same shared primary key pattern as the `student` table. It stores the `Designation` (referencing the `lookup` table) and the `Salary` of each advisor. The salary is stored as a `DECIMAL(18, 0)` to handle large values without floating-point precision errors. The `Designation` column enforces referential integrity with the `lookup` table, meaning it is impossible to assign a designation that does not exist in the predefined list.

3.1.5 Project Table

The `project` table stores FYP project titles and their full descriptions. The `Title` column is limited to 50 characters to keep project names concise, while the `Description` column is defined as `LONGTEXT` to allow arbitrarily detailed explanations of the project's scope, objectives, and expected outcomes. Every project is uniquely identified by an auto-incremented `Id`.

3.1.6 Group Table

The `group` table stores student groups. Each group is assigned an auto-incremented `Id` and the date on which it was created, recorded in the `Created_On` column. The creation date is inserted automatically by the application using the system's current date at the time the group is formed. The group `Id` is then used as a foreign key in the `groupstudent`, `groupproject`, and `groupevaluation` tables, making it the central linking point for all group-related information.

3.1.7 GroupStudent Table

The `groupstudent` table is a junction table that models the many-to-many relationship between groups and students. A group can contain many students, and in principle a student's history across multiple groups can be tracked over time. The composite primary key `(GroupId, StudentId)` ensures that the same student cannot be added to the same group twice. Each membership record also carries a `Status` (Active or Inactive, from the `lookup` table) and an `AssignmentDate` that records when the student was added.

3.1.8 GroupProject Table

The `groupproject` table links a project to a group. Each group is assigned exactly one project, and once a project has been assigned it is no longer shown as available for other groups. The composite primary key `(ProjectId, GroupId)` prevents the same project from being assigned to the same group more than once. The `AssignmentDate` column records when the project was formally assigned, providing an audit trail that the committee can refer to if disputes arise.

3.1.9 ProjectAdvisor Table

The `projectadvisor` table links advisors to a project, with each link carrying a defined `AdvisorRole` (Main Advisor, Co-Advisor, or Industry Advisor). The composite primary key `(AdvisorId, ProjectId)` prevents the same advisor from being assigned to the same project in any role more than once. The `AdvisorRole` column references the `lookup` table, ensuring only valid roles can be assigned. The `AssignmentDate` records when the advisor was formally assigned to the project.

3.1.10 Evaluation Table

The `evaluation` table stores the types of evaluations conducted during the FYP cycle. Examples include Proposal Defense, Mid Evaluation, and Final Evaluation. Each evaluation type has a `Name`, a `TotalMarks` value (the maximum marks a group can score), and a `TotalWeightage` value (the percentage contribution of that evaluation to the final grade). By storing these values in a dedicated table rather than hard-coding them, the committee can define any number of evaluation rounds without requiring changes to the application.

3.1.11 GroupEvaluation Table

The `groupevaluation` table stores the actual marks obtained by each group in each evaluation. The composite primary key `(GroupId, EvaluationId)` ensures that a group can be evaluated in any given evaluation type only once. The `ObtainedMarks` column stores the raw score, and the `EvaluationDate` records when the evaluation took place. When displaying results, the application calculates the weighted score on the fly using the formula `(ObtainedMarks / TotalMarks) * TotalWeightage`, which gives the score as a percentage of the evaluation's overall weight in the curriculum.

4. Application Modules

4.1 Overview

The application consists of the following eight modules, each implemented as a dedicated Windows Form. The modules are accessible from a central navigation menu on the main screen, allowing committee members to switch between tasks without closing and reopening the application.

1. Student Management
2. Advisor Management
3. Project Management
4. Group Management
5. Assign Student to Group
6. Assign Project to Group
7. Assign Advisor to Project
8. Evaluation Management

4.2 Student Management

4.2.1 What It Does

This screen allows the committee to add new students to the system, view all existing students in a data grid, edit the information of any selected student, and delete records that are no longer needed. Because the student's data is split across the `person` and `student` tables, every add or edit operation involves two SQL statements executed in the correct order to respect the foreign key relationship. The data grid displays all student information in a single combined view using a `JOIN` query.

4.2.2 Fields on the Form

First Name, Last Name, Contact Number, Email, Date of Birth, Gender (dropdown populated from `lookup`), Registration Number.

4.2.3 Screenshot

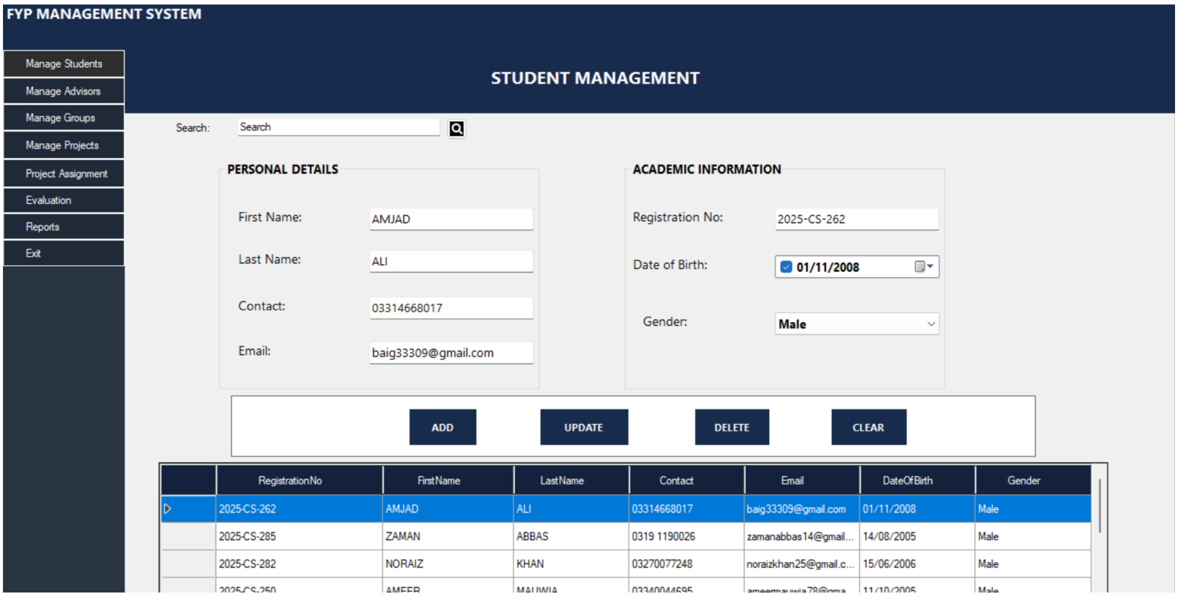


Figure 4.1: Student Management Form

4.3 Advisor Management

4.3.1 What It Does

This screen allows the committee to add, view, edit, and delete advisor records. Since an advisor is also a person, every add or update operation involves writing to both the `person` and `advisor` tables.

4.3.2 Screenshot

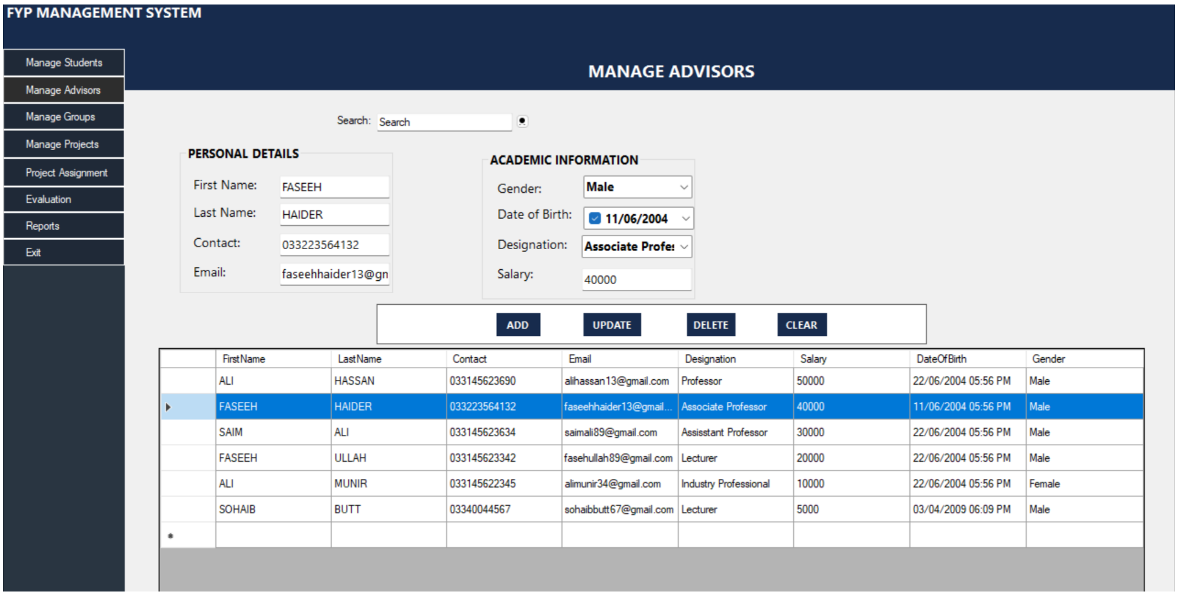


Figure 4.2: Advisor Management Form

4.4 Project Management

4.4.1 What It Does

This screen allows the committee to add project titles with full descriptions, update existing project details, and delete projects. A particularly useful feature of this screen is the assignment status column, which dynamically indicates whether each project has already been assigned to a group or is still available. This is determined at query time by performing a `LEFT JOIN` with the `groupproject` table and using a `CASE` expression to produce a readable label. The committee can therefore see at a glance how many projects remain unassigned without needing to cross-reference a separate list.

4.4.2 Fields on the Form

Project Title, Description.

4.4.3 Screenshot

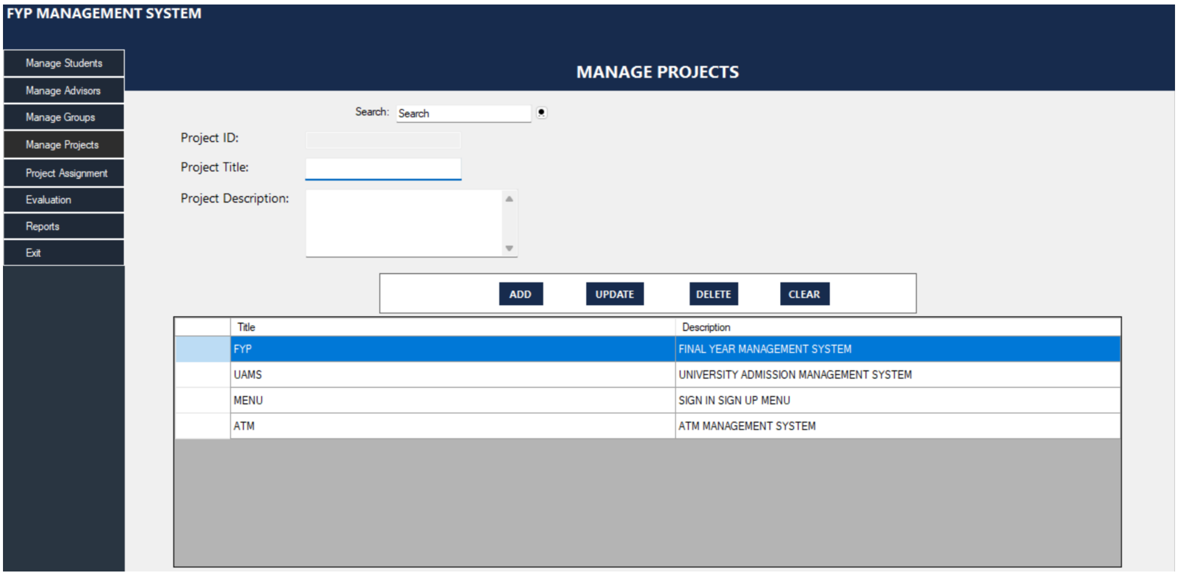


Figure 4.3: Project Management Form

4.5 Group Management

4.5.1 What It Does

This screen allows the committee to create a new student group. When a group is created, the current date is saved automatically as its creation date. Once a group exists, students can be added to it through a sub-form that lists all registered students.

4.5.2 Screenshot

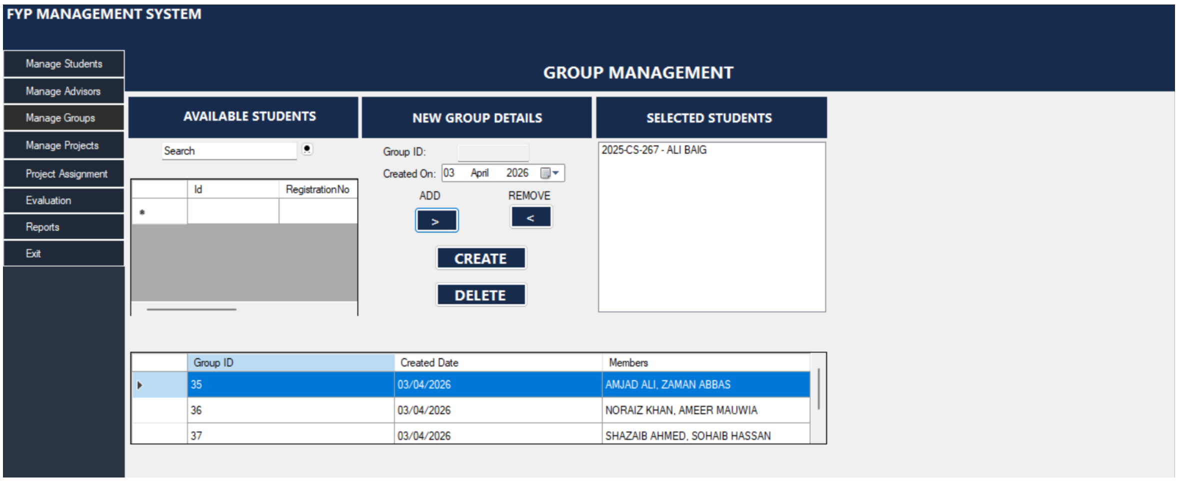


Figure 4.4: Group Management Form

4.6 Assign Project and Advisors

4.6.1 What It Does

This part allows the committee to select an existing group and assign an available project to it. Once the assignment is confirmed, the record is saved in the `groupproject` table with the current date as the assignment date.This part allows the committee to assign advisors to a specific project. Three advisor roles are supported: Main Advisor, Co-Advisor, and Industry Advisor, each selected from a dropdown populated from the `lookup` table. A separate advisor can be assigned in each role, or the same advisor can theoretically hold multiple roles if the department's circumstances require it.

4.6.2 Screenshot

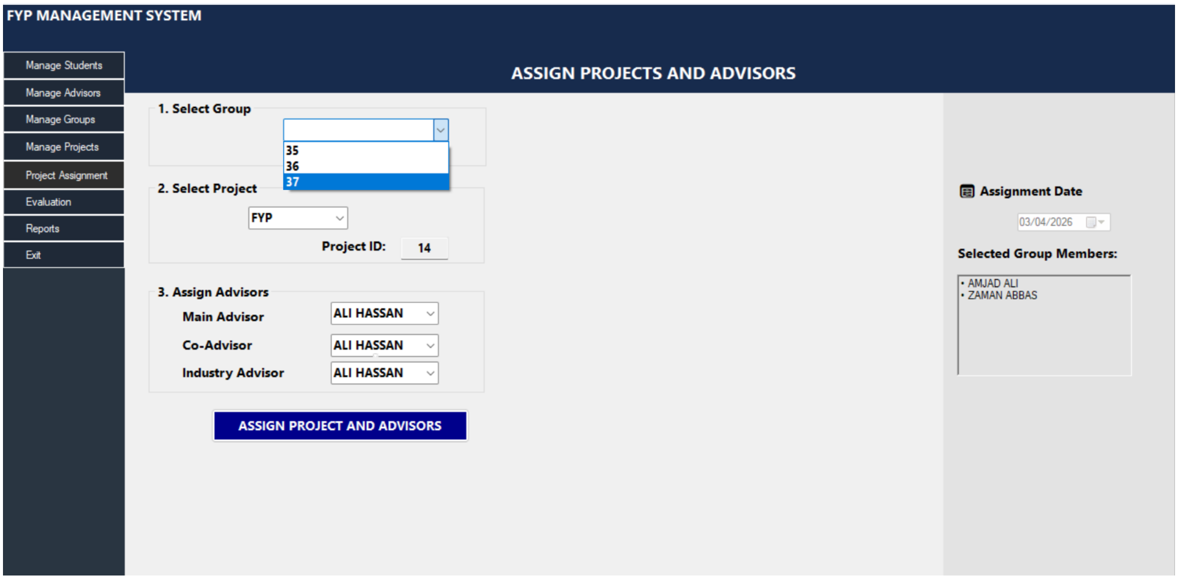


Figure 4.5: Assign Project and Advisors Form

4.7 Evaluation Management

4.7.1 What It Does

This module has two distinct sub-parts. The first allows the committee to define new evaluation types — for example, “Mid Examination”, “Final Report”, or “Final Viva” — specifying the total marks available and the percentage weightage that evaluation contributes to the overall FYP grade.

The second part allows the committee to record the actual marks obtained by each group in a selected evaluation.

4.7.2 Fields for Evaluation Type

Evaluation Name, Total Marks, Total Weightage (percentage).

4.7.3 Fields for Recording Marks

Select Group, Select Evaluation, Obtained Marks, Evaluation Date.

4.7.4 Screenshot

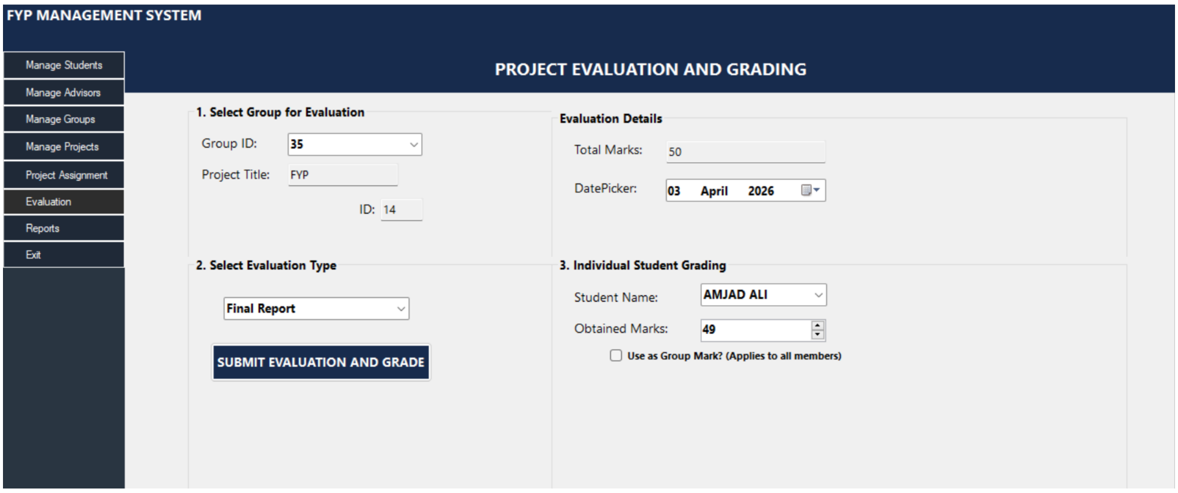


Figure 4.6: Evaluation Management Form

4.8 Project Report Management

4.8.1 What It Does

This module serves as the central reporting hub of the FYP Management System, allowing the committee to generate and export structured summaries of all project-related data from a single unified interface. The module is divided into two distinct sub-parts corresponding to the two supported report types: the **Project Advisor Allocation List** and the **Student Marks Sheet**.

4.8.2 Report Configuration Fields

Report Type (dropdown: Student Marks Sheet / Project Advisor Allocation List), Academic Year (dropdown), Evaluation Type Filter (optional dropdown).

4.8.3 Filtering Options Fields

Project Status (dropdown), Date From (date picker), Date To (date picker).

4.8.4 Screenshot

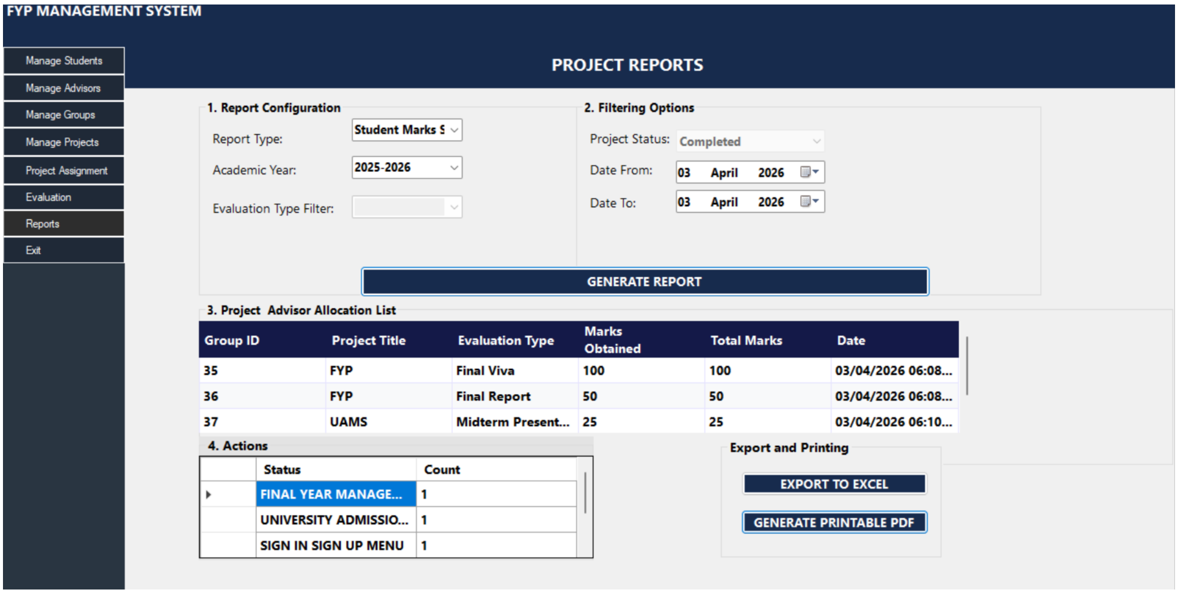


Figure 4.7: Report Management Form

5. PDF Reports

5.1 Overview

The application can generate two types of reports on demand, each available in two export formats: a printable PDF and an Excel spreadsheet. When the committee member selects a report type and clicks the relevant export button, they are prompted to choose a save location using a standard file save dialog.

5.2 Report 1: Project Advisor Allocation – PDF

5.2.1 What It Does

This report generates a printable PDF document providing the committee with a complete overview of all FYP projects, their assigned advisors (with roles), and the students belonging to each group.

5.2.2 Screenshot

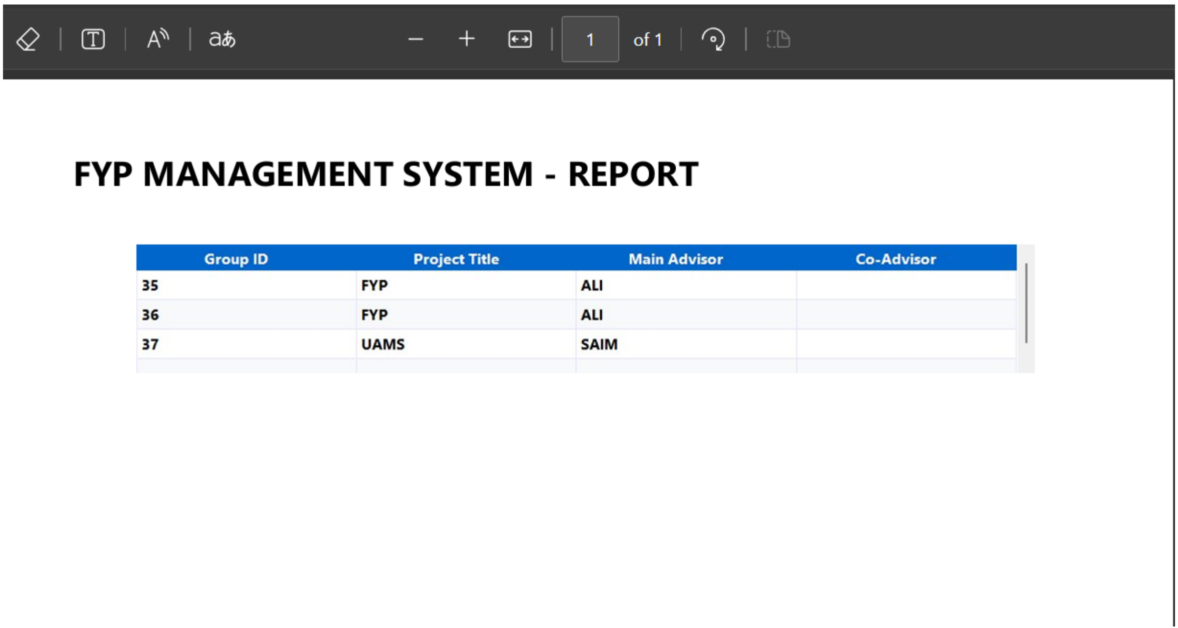


Figure 5.1: Report 1 – Project Advisor Allocation PDF Export

5.3 Report 2: Project Advisor Allocation – Excel

5.3.1 What It Does

This report exports the same Project Advisor Allocation data to a `.xlsx` Excel file by clicking **Export to Excel**.

5.3.2 Screenshot

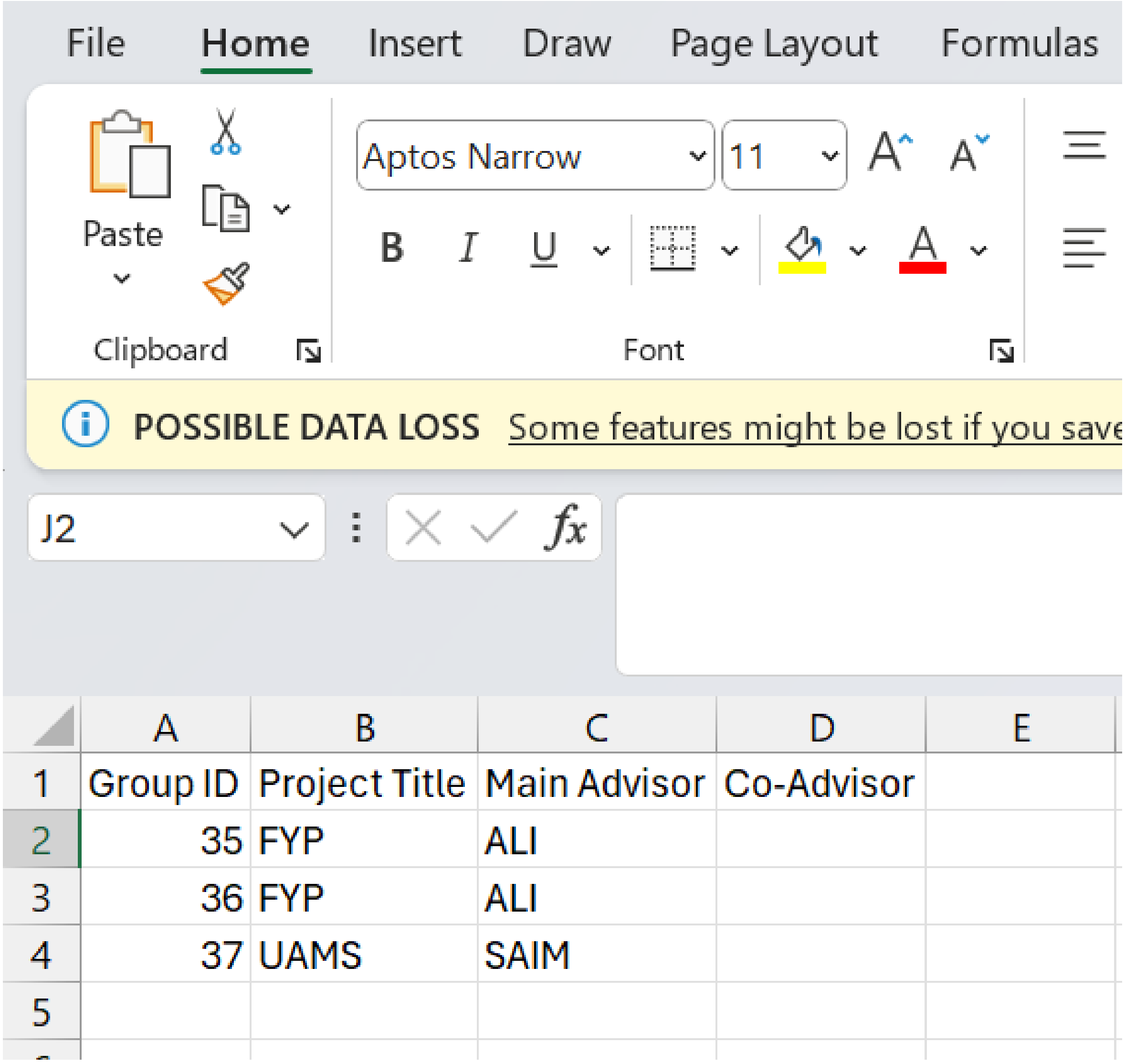


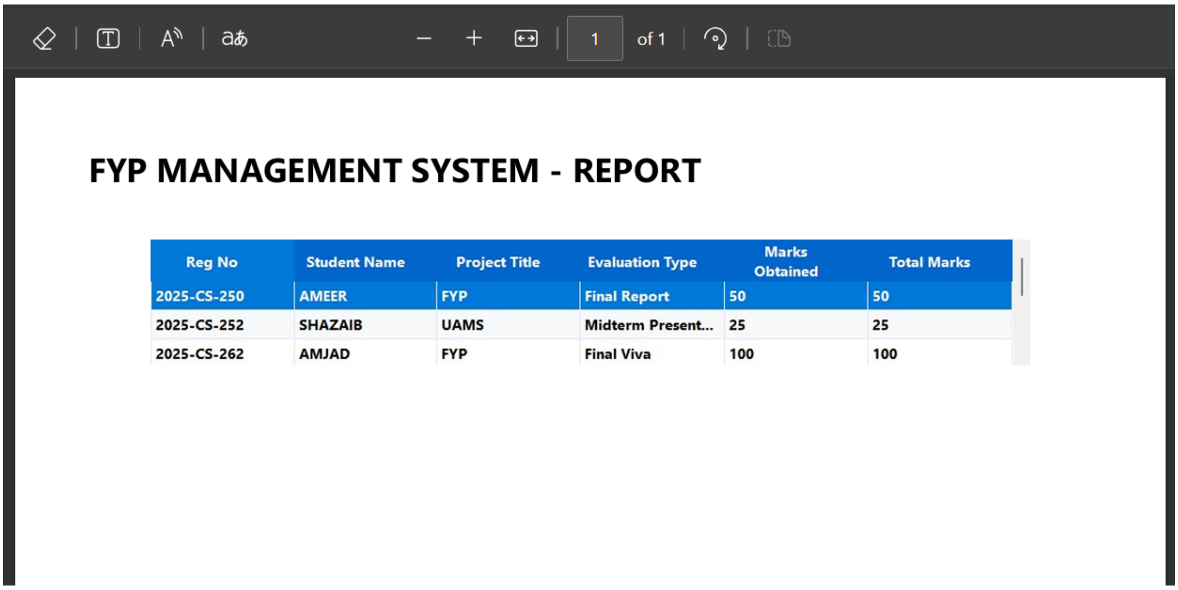
Figure 5.2: Report 2 – Project Advisor Allocation Excel Export

5.4 Report 3: Student Marks Sheet – PDF

5.4.1 What It Does

This report generates a printable PDF marks sheet displaying the evaluation performance of every group. For each group it lists the students by name and registration number.

5.4.2 Screenshot



FYP MANAGEMENT SYSTEM - REPORT

Reg No	Student Name	Project Title	Evaluation Type	Marks Obtained	Total Marks
2025-CS-250	AMEER	FYP	Final Report	50	50
2025-CS-252	SHAZAIB	UAMS	Midterm Present...	25	25
2025-CS-262	AMJAD	FYP	Final Viva	100	100

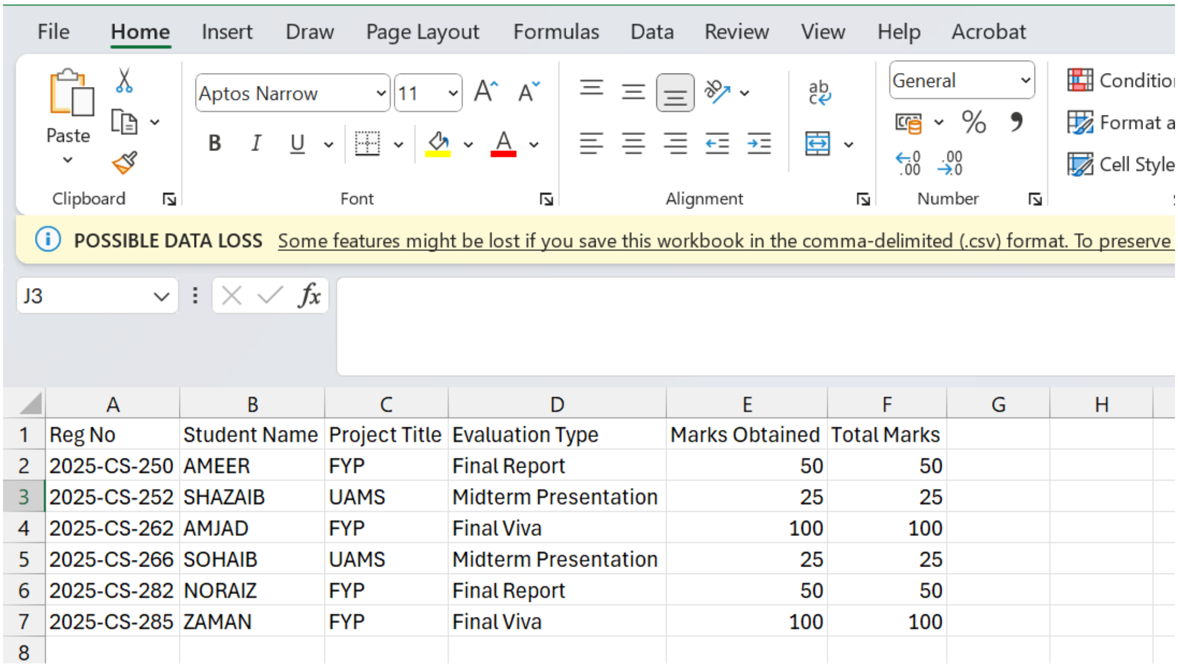
Figure 5.3: Report 3 – Student Marks Sheet PDF Export

5.5 Report 4: Student Marks Sheet – Excel

5.5.1 What It Does

This report exports the Student Marks Sheet data to a `.xlsx` Excel file. The file is generated instantly after clicking **Export to Excel** with the Student Marks Sheet report type selected.

5.5.2 Screenshot



	A	B	C	D	E	F	G	H
1	Reg No	Student Name	Project Title	Evaluation Type	Marks Obtained	Total Marks		
2	2025-CS-250	AMEER	FYP	Final Report	50	50		
3	2025-CS-252	SHAZAIB	UAMS	Midterm Presentation	25	25		
4	2025-CS-262	AMJAD	FYP	Final Viva	100	100		
5	2025-CS-266	SOHAIB	UAMS	Midterm Presentation	25	25		
6	2025-CS-282	NORAIZ	FYP	Final Report	50	50		
7	2025-CS-285	ZAMAN	FYP	Final Viva	100	100		
8								

Figure 5.4: Report 4 – Student Marks Sheet Excel Export

6. Conclusion

This project delivered a fully functional desktop application — the **FYP Management System** — developed as the course project for CS104 Database Systems. The application enables the FYP committee at UET Lahore to manage students, advisors, projects, groups, and evaluations from a single unified interface, replacing the previous error-prone Excel-based workflow with a structured, reliable, and database-backed solution.

The application was built using C# Windows Forms and communicates with a MySQL database exclusively through hand-written, parameterised SQL queries, as required by the course rules. The provided database schema was not modified in any way. Data integrity is enforced at the database level through primary keys, foreign keys, and composite key constraints, while additional validation at the application level prevents common input mistakes from reaching the database at all. Errors are handled gracefully throughout the application, with informative message boxes displayed to guide the user rather than allowing the application to crash.

Working on this project provided valuable, hands-on experience in several areas. Writing `JOIN` queries across multiple related tables — including the shared primary key pattern used for the `person`, `student`, and `advisor` tables — deepened the understanding of how relational databases model real-world entities. Designing the Data Access Layer as a separate module demonstrated the importance of separating concerns in software development: keeping SQL out of the UI code makes the application easier to maintain, test, and extend. The use of the `lookup` table as a centralised reference for all categorical values also illustrated how a well-designed schema can reduce redundancy and improve consistency across the entire system.

Future improvements to this system could include adding a user login screen so that different committee members can have their own accounts, implementing a full audit log to track which records were changed and by whom, and extending the reporting module to support additional report formats or more detailed analytical views of evaluation results across multiple semesters.